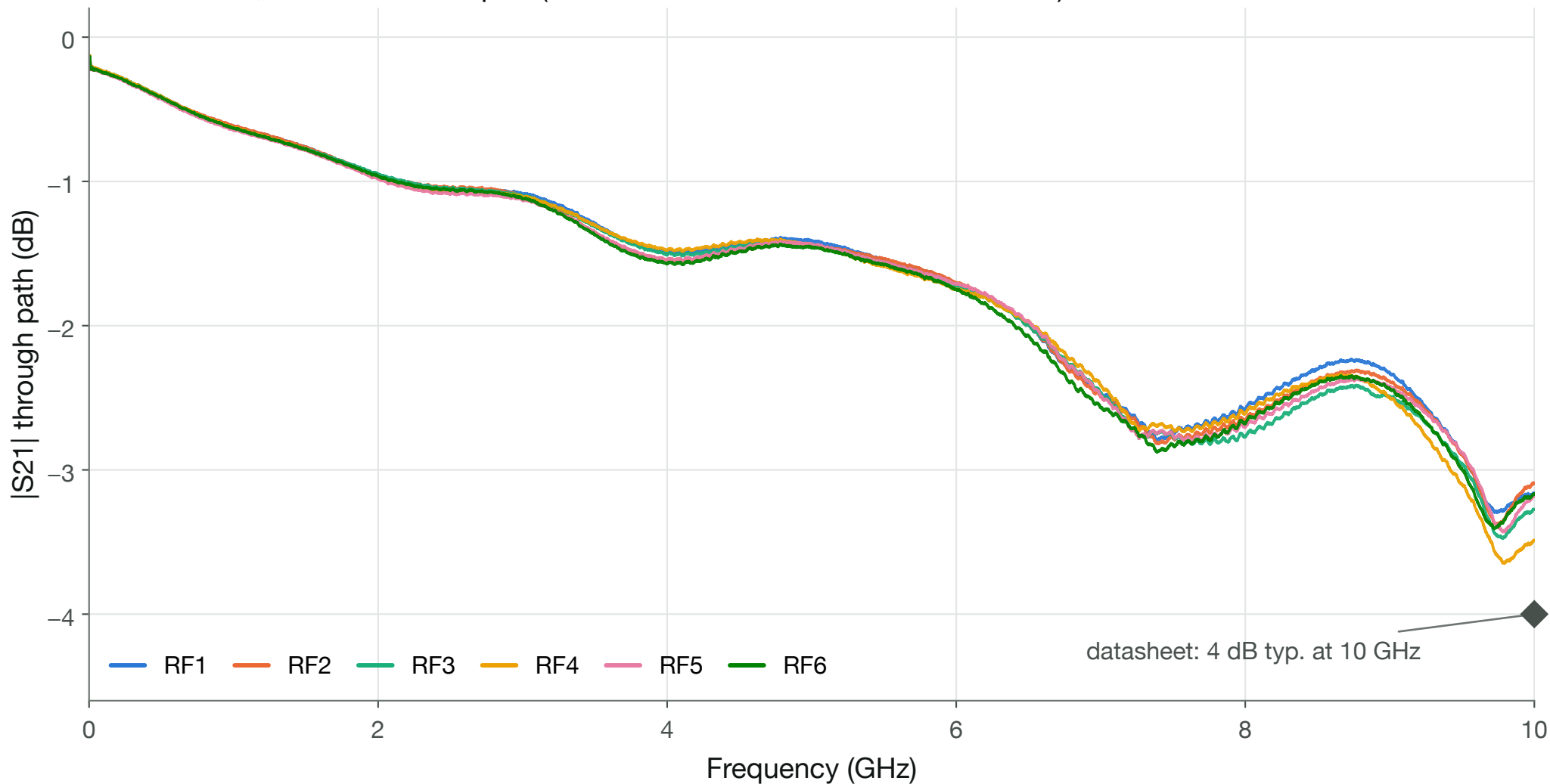
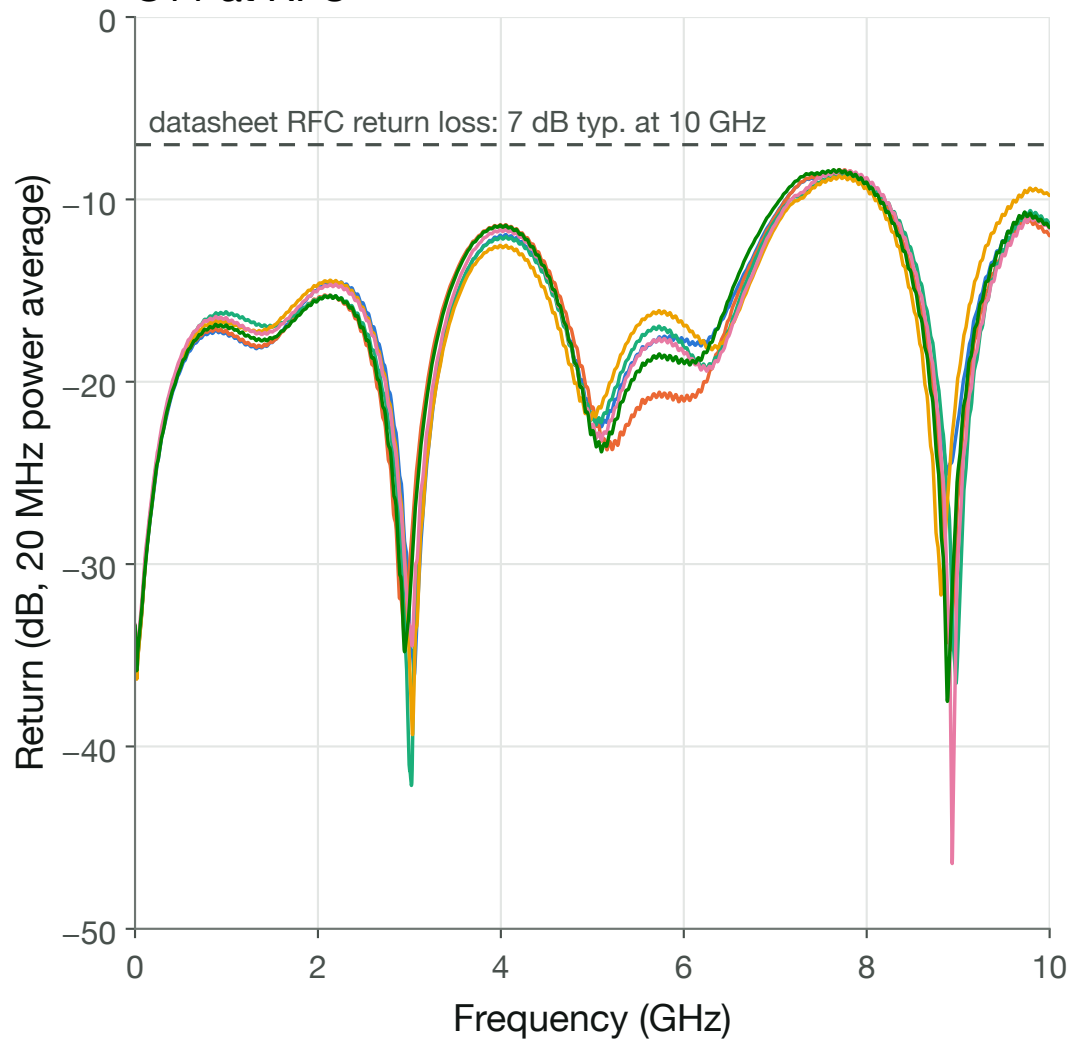


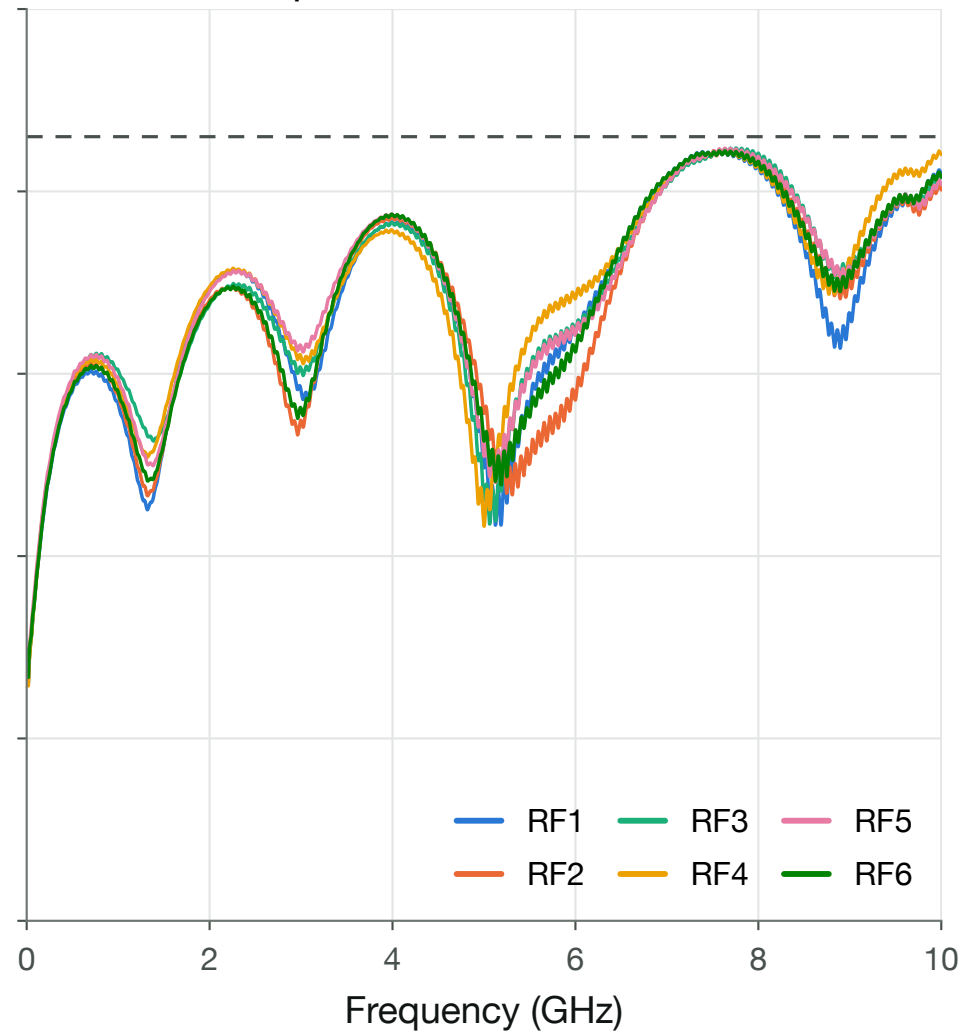
Insertion loss, RFC to each RF port (VNA-calibrated at the switch connectors)



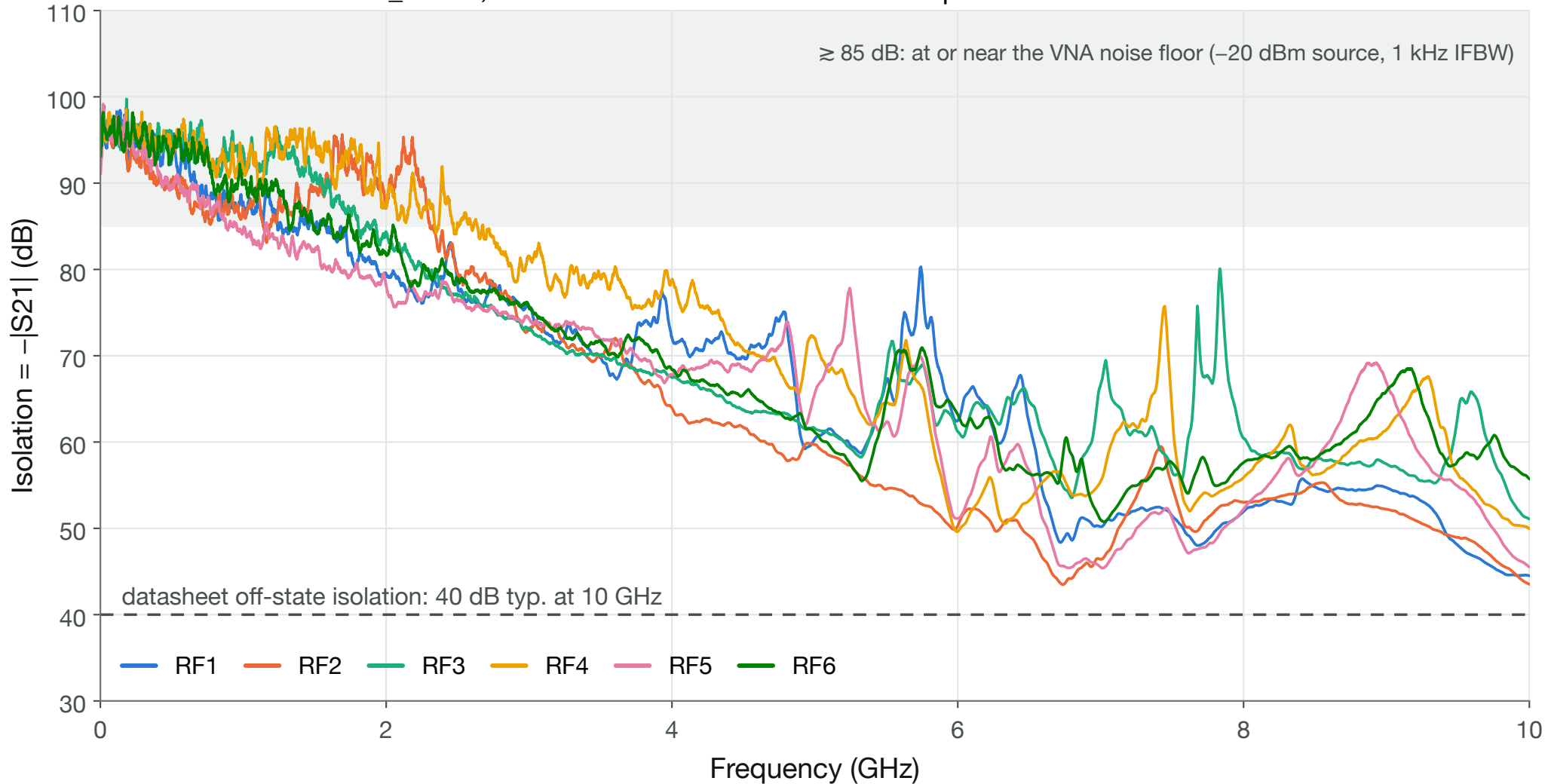
S11 at RFC



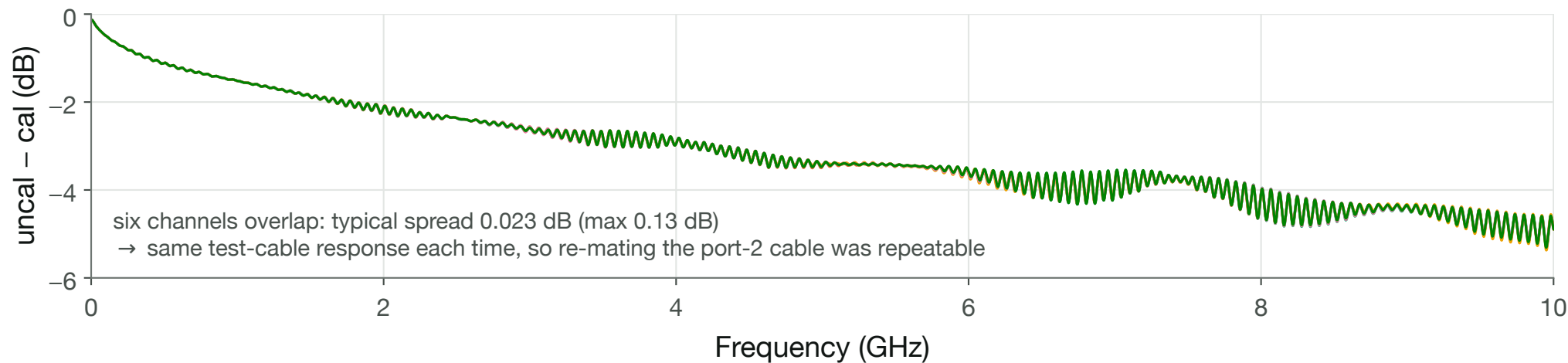
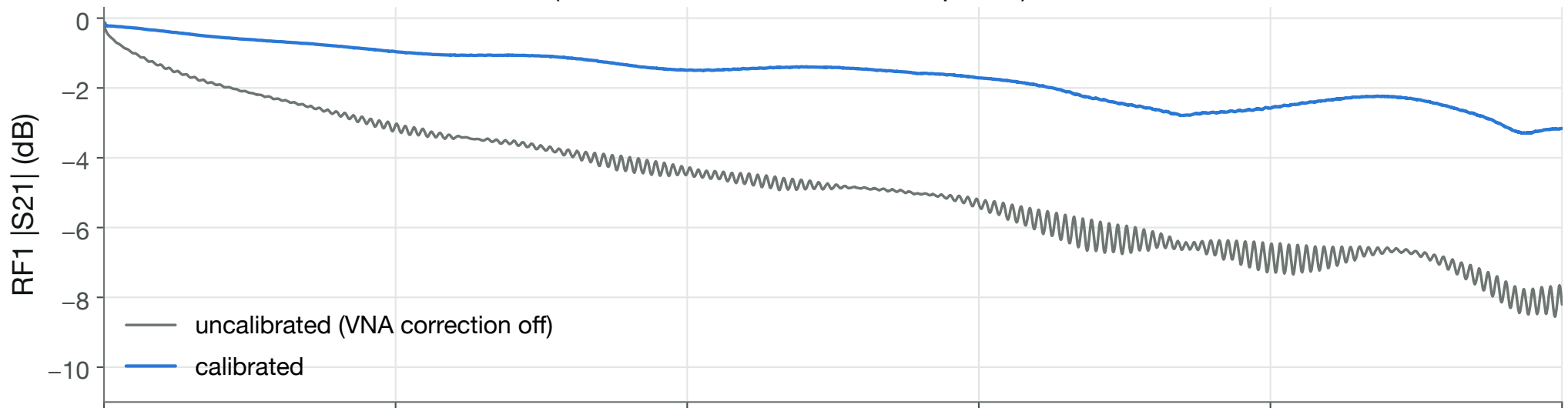
S22 at the RF port



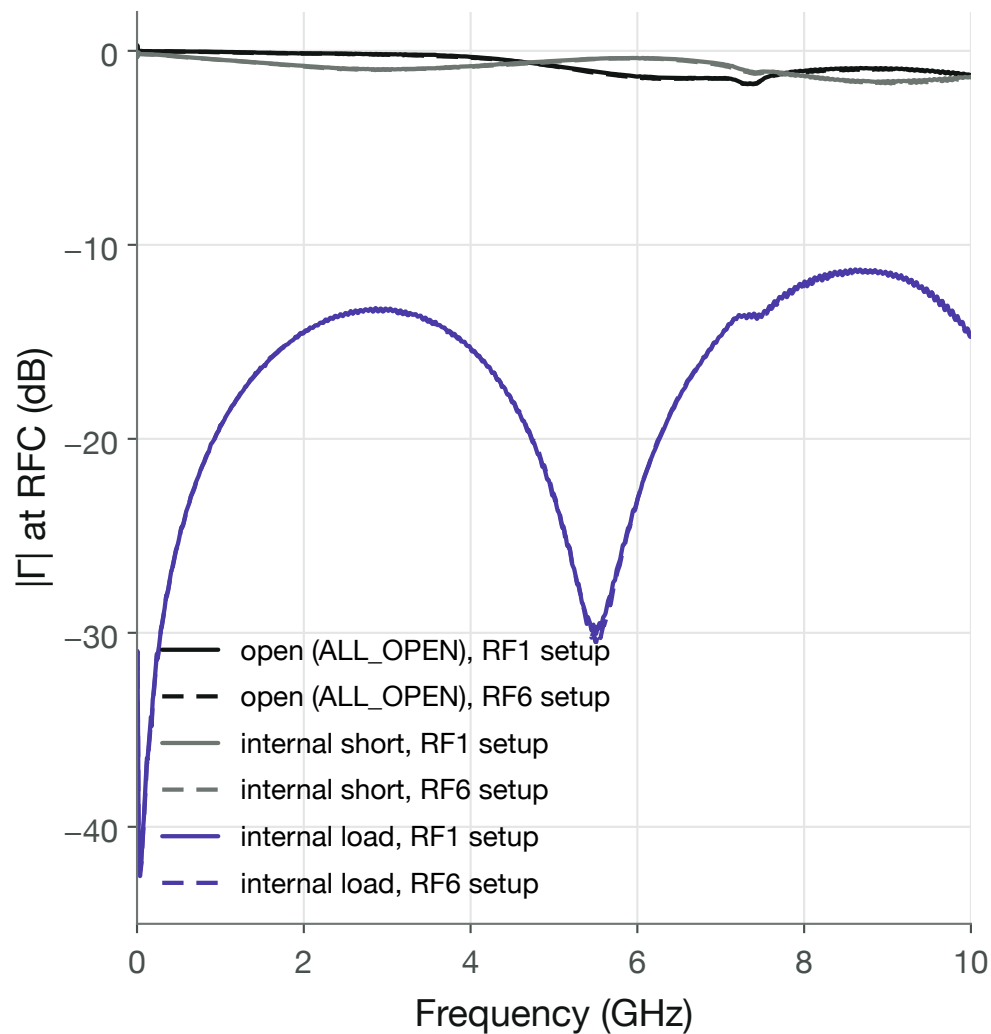
Off-state isolation: ALL_OPEN, VNA on RFC and on the named RF port



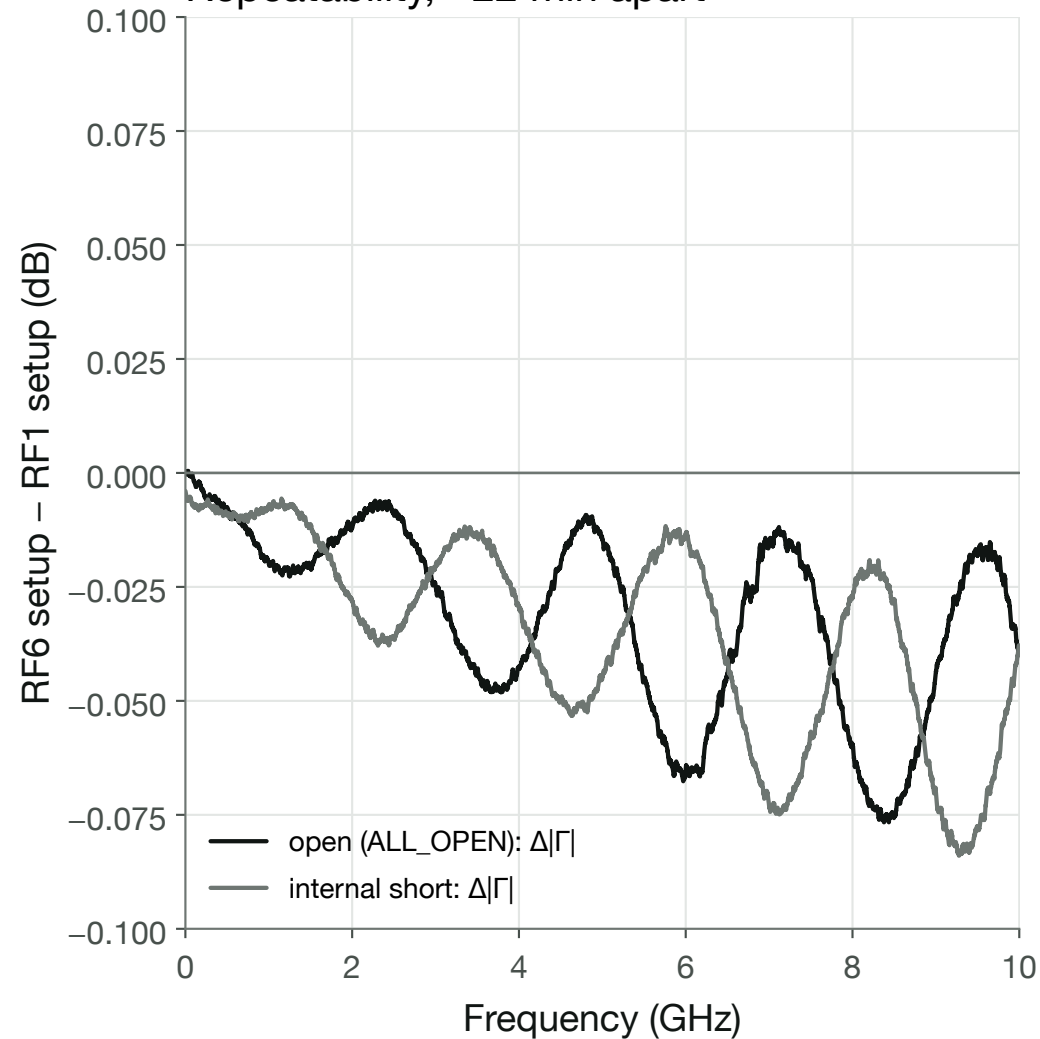
What the VNA calibration removes (test cables + VNA test-set response)



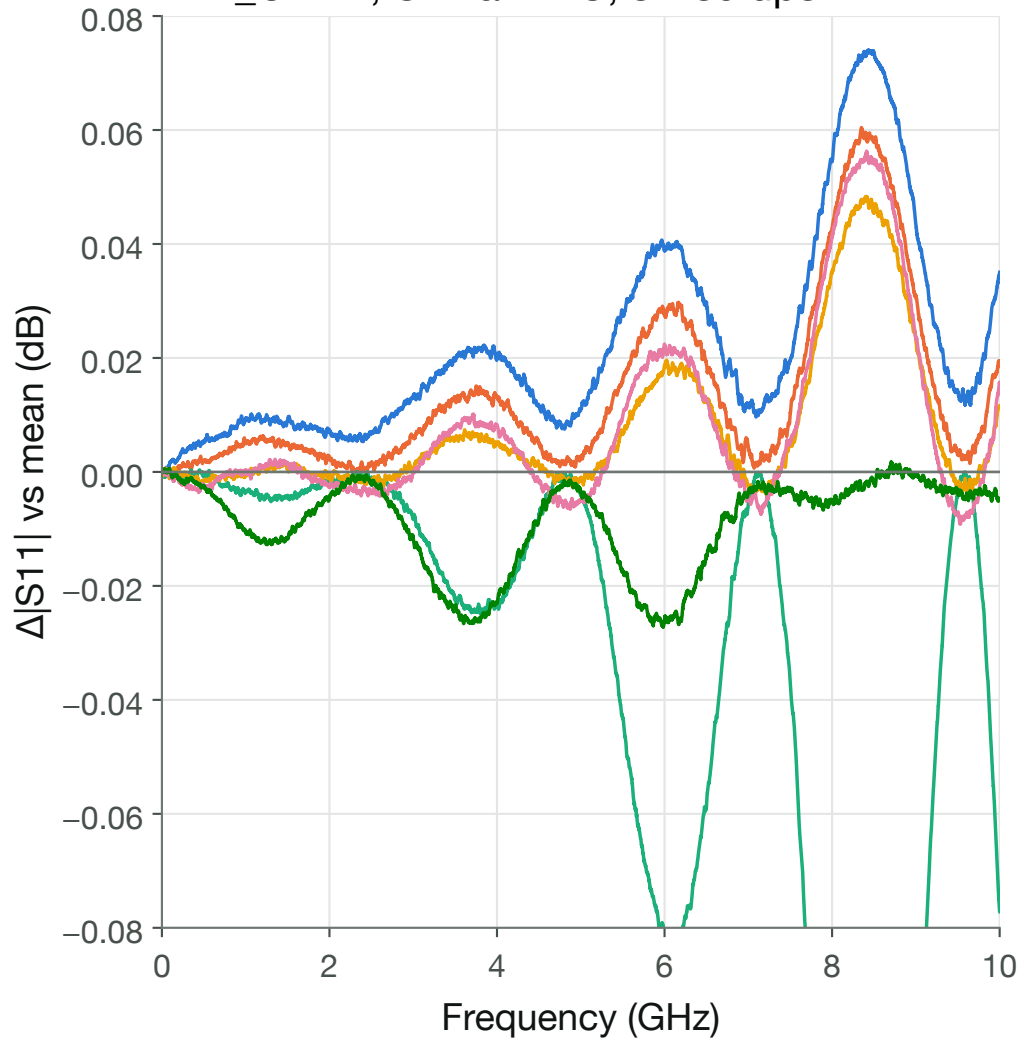
Internal standards seen at RFC



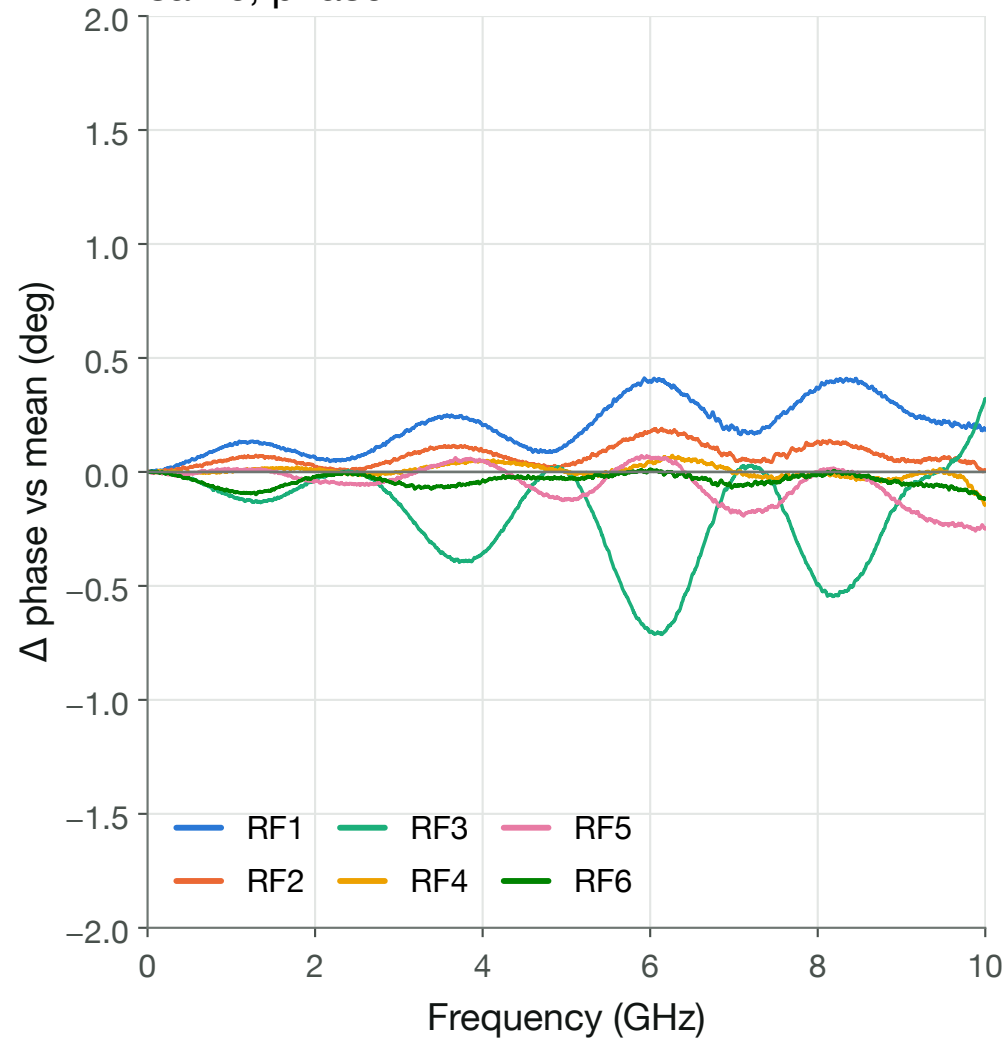
Repeatability, ~22 min apart

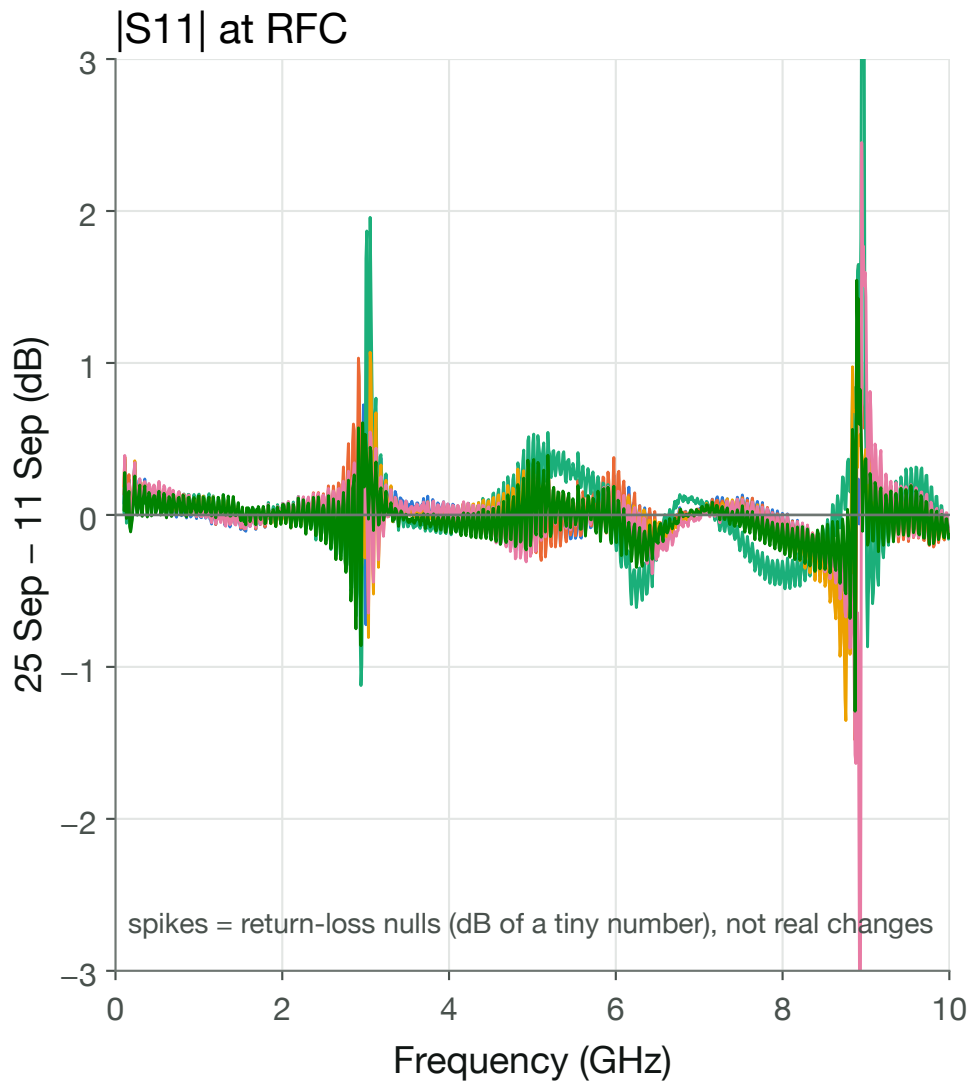
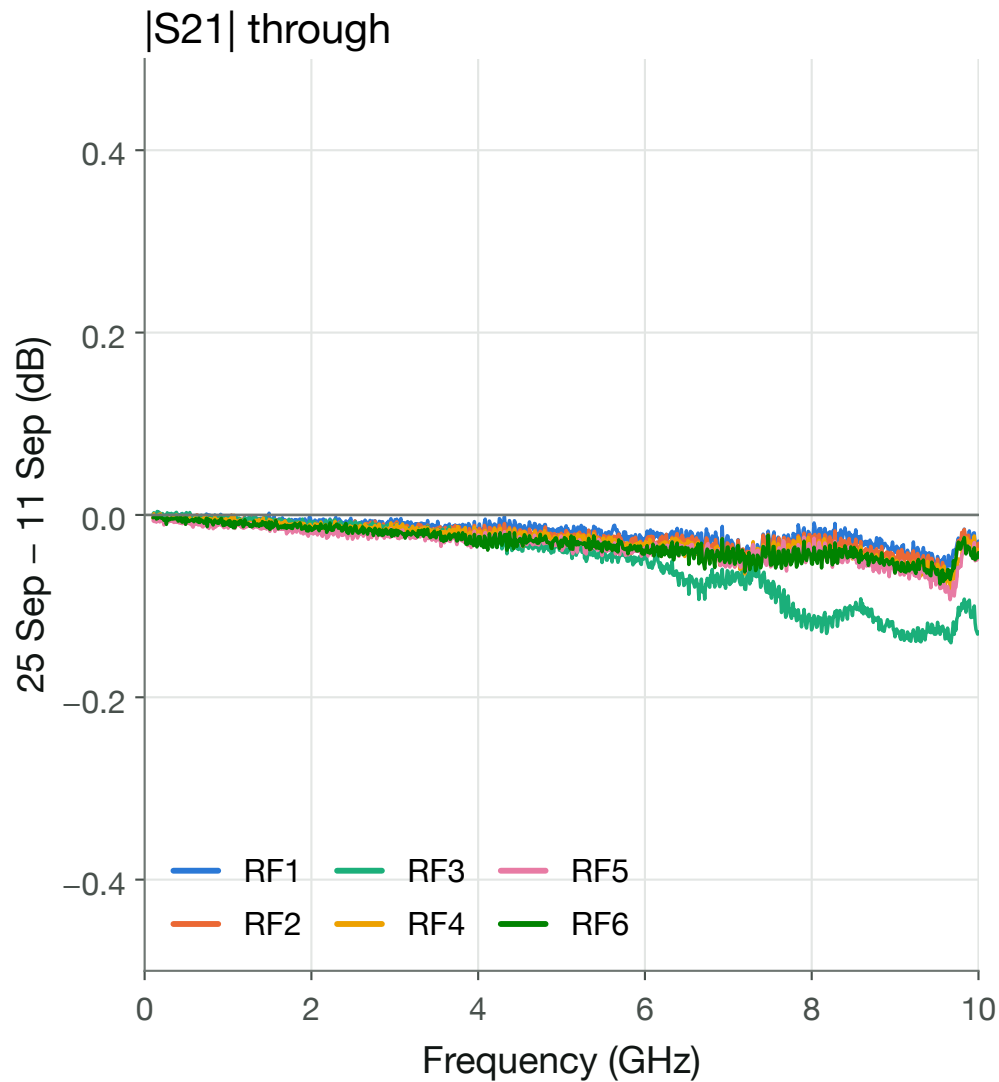


ALL_OPEN, S11 at RFC, six setups

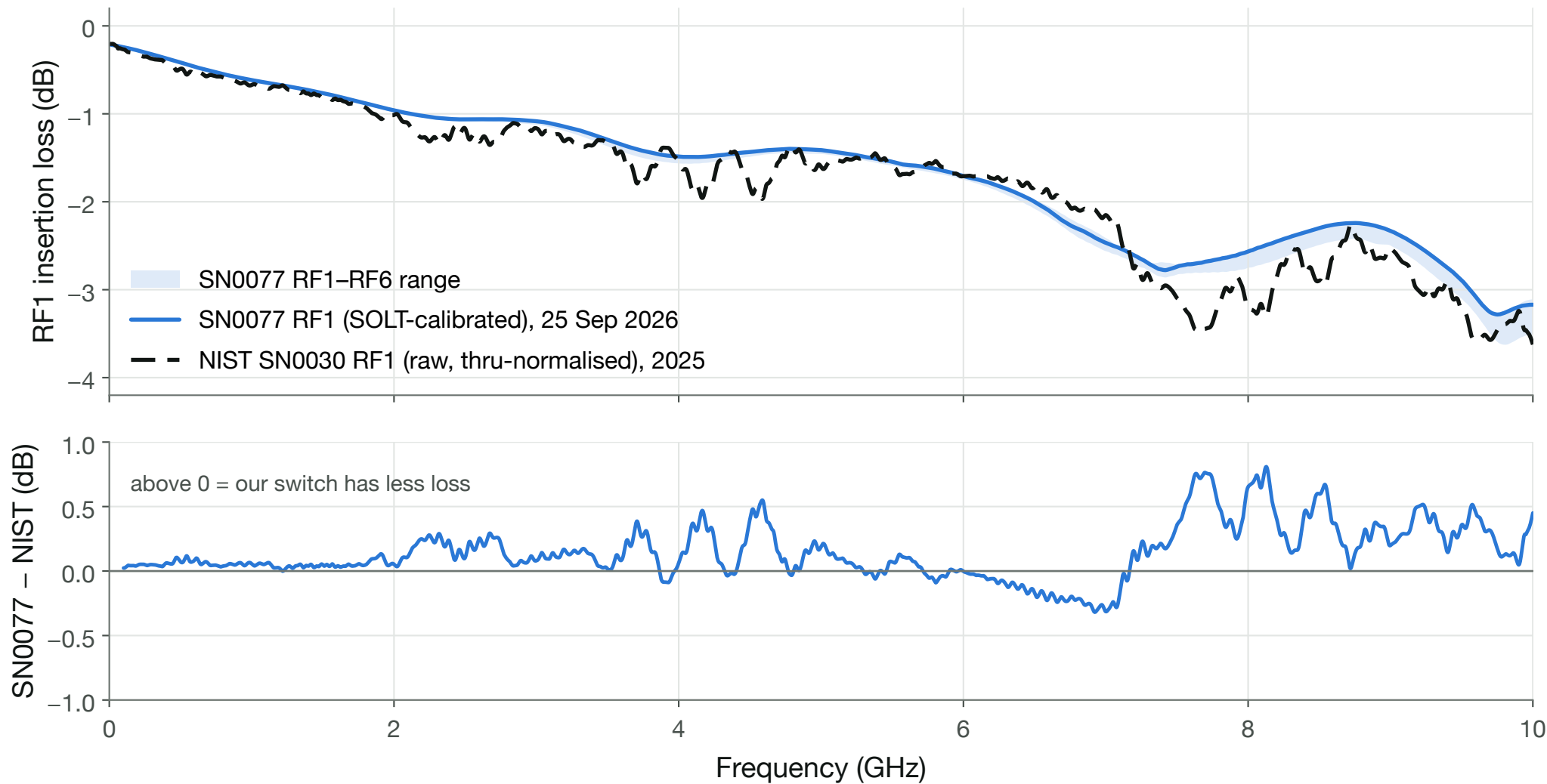


same, phase





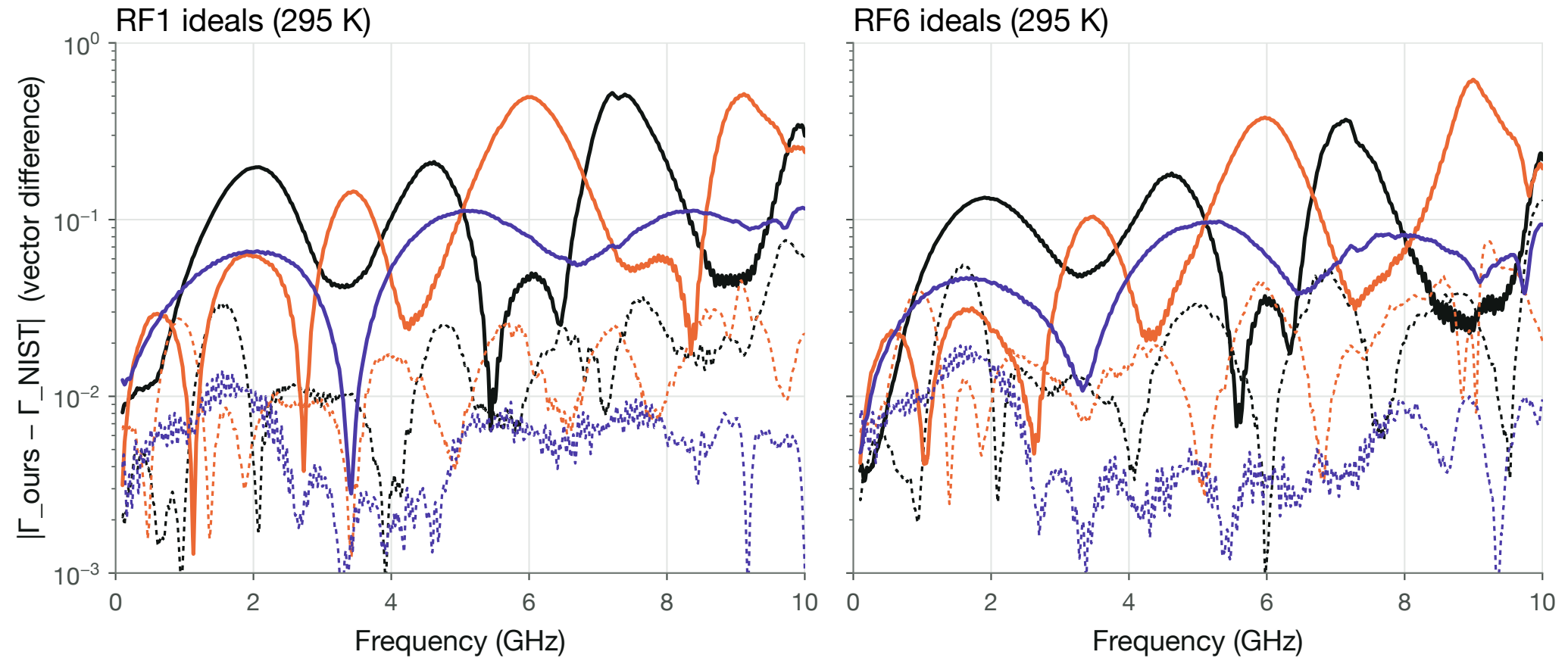
Insertion loss vs NIST at 295 K: both referenced to the switch SMA connectors



Both curves 100 MHz running mean (NIST's thru-normalisation leaves mismatch ripple a SOLT cal removes) · NIST thru repeats agree to 0.01 dB

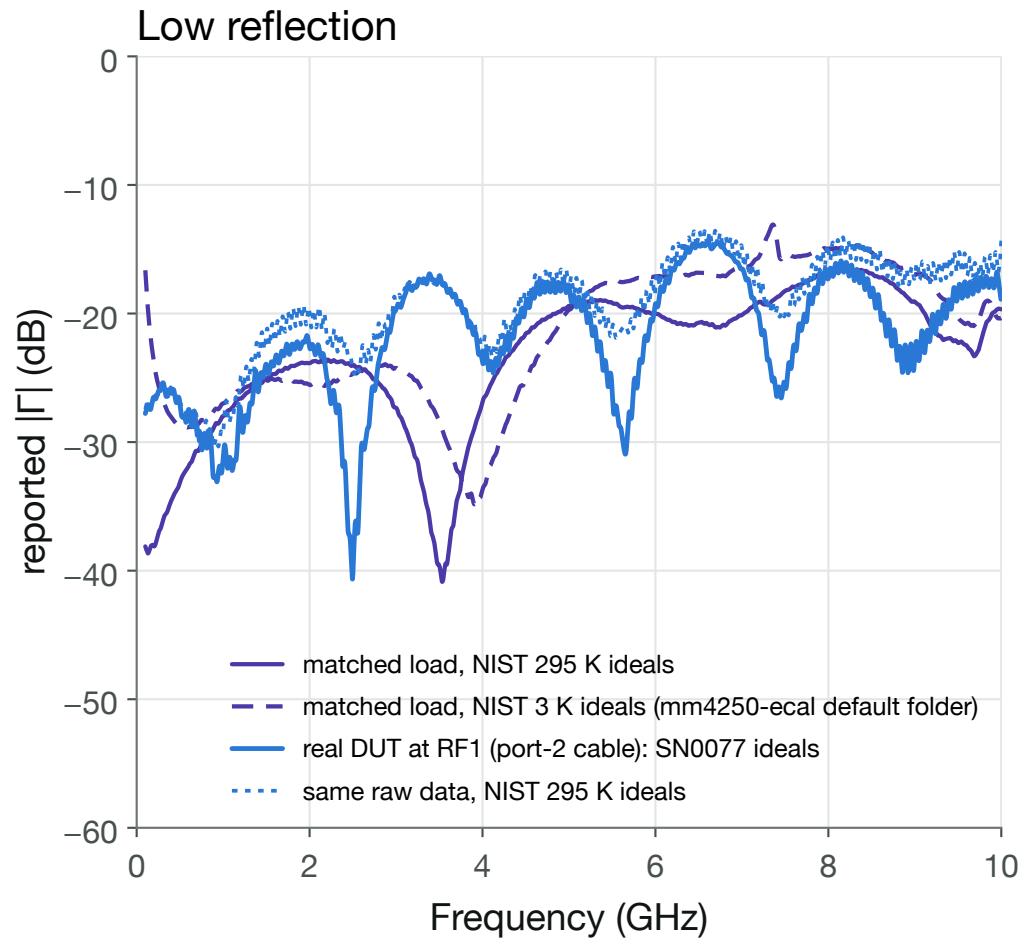
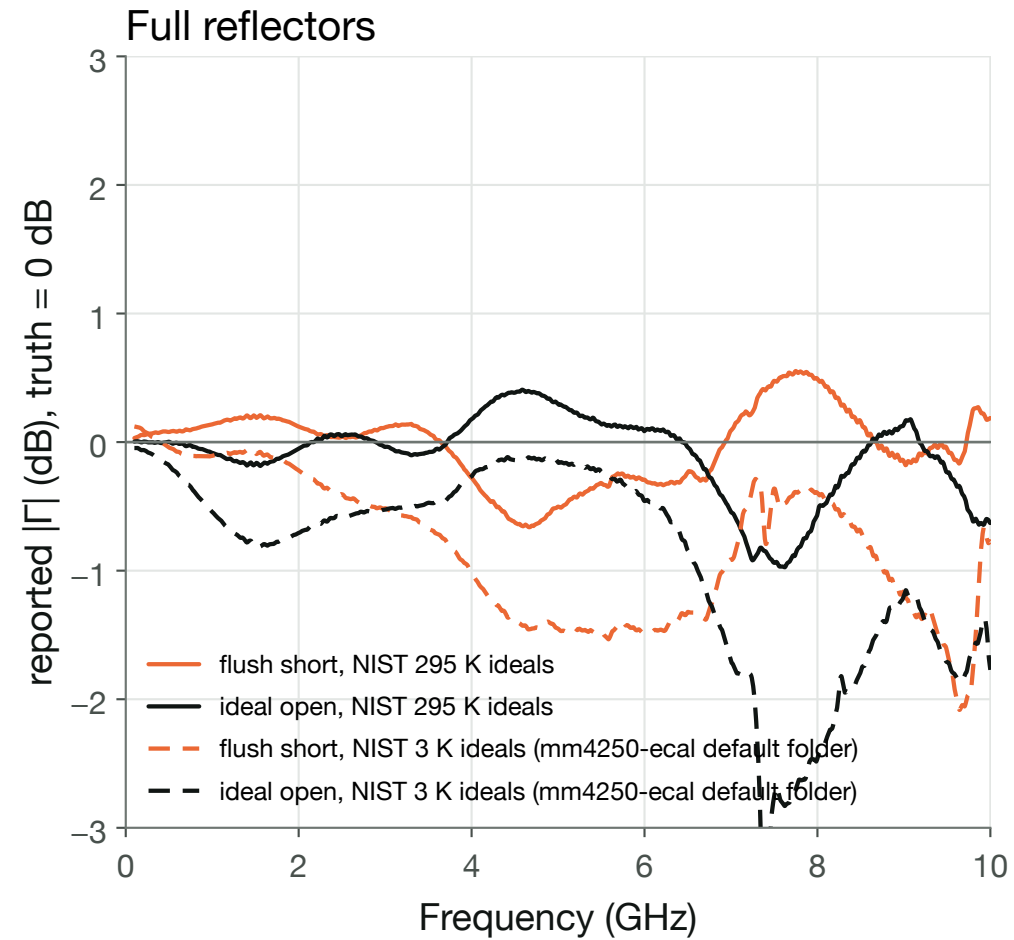
Internal-standard definitions at the RF-port SMA plane: our SOLT-derived vs NIST's 295 K ideals

— open — short — load — SN0077 vs NIST SN0031 ---- NIST cal 1 vs cal 2, same switch (NIST's own repeatability)



Ours: calibrated standard at RFC with the calibrated RFC → RFn path de-embedded · 50 MHz running mean · 0.01 $\approx$ -40 dB error vector

E-cal on SN0077 RF1 with NIST's generic ideals: the error you would get



Truth = SN0077 ideals derived from the SOLT data · 50 MHz running mean